Machine-Learning Reproducibility Details

This document records the implementation parameters required to reproduce the machine-learning experiments. The attack methodology, trace representations, GE/SR definitions, and countermeasure constructions are described in the accompanying paper and are not repeated here.

For compactness, define

$$\mathcal{D} = \{d_{\text{abs}}, d_{\text{minus}}, d_{\text{plus}}, d_{\text{mul}}\}, \qquad \mathcal{D}_{\text{aux}} = \{d_{\text{minus}}, d_{\text{plus}}, d_{\text{mul}}\}.$$

Architecture keys. For compactness, we refer to the following model backbones throughout this document.

- **A1 AESPTv2 CNN:** Conv2D$(64, 3\times1)$ $\to$ ReLU $\to$ Dropout(0.2) $\to$ AvgPool(2×1) $\to$ Conv2D$(128, 3\times1)$ $\to$ ReLU $\to$ Dropout(0.1) $\to$ AvgPool(2×1) $\to$ Conv2D$(128, 3\times1)$ $\to$ ReLU $\to$ Dropout(0.1) $\to$ AvgPool(2×1) $\to$ Dense(32) $\to$ ReLU $\to$ Dropout(0.1).
- **A2 AESPTv2 LSTM:** LSTM$(64, \text{sequence})$ $\to$ Dropout(0.4) $\to$ LSTM$(64, \text{sequence})$ $\to$ Dropout(0.2) $\to$ LSTM$(32, \text{last})$.
- **A3(C) ARM CNN with C input channels:** Conv1D$(64, 3, \text{ReLU}, \text{same})$ $\to$ Dropout(0.2) $\to$ AvgPool(2) $\to$ Conv1D$(128, 3, \text{ReLU}, \text{same})$ $\to$ Dropout(0.1) $\to$ AvgPool(2) $\to$ Conv1D$(128, 3, \text{ReLU}, \text{same})$ $\to$ Dropout(0.1) $\to$ AvgPool(2) $\to$ Flatten $\to$ Dense$(32, \text{ReLU})$ $\to$ Dropout(0.1).
- **A4 XOR-aware ARM CNN:** same convolutional stack as **A3(8)**, but with Dense$(64, \text{ReLU})$ $\to$ Dropout(0.1) before the output layer.

Input keys. For compactness, we use the following input configurations:

$$\begin{aligned}
\mathcal{I}_1 &= \{d_\star(P_i) : d_\star \in \mathcal{D}\},\\
\mathcal{I}_{\text{intra}} &= \{[d_{\text{abs}}(P_i), d_\star(P_i)] : d_\star \in \mathcal{D}_{\text{aux}}\},\\
\mathcal{I}_{\text{raw}} &= [x_i^{(1)}, x_i^{(2)}],\\
\mathcal{I}_{\text{inter}} &= \{[d_\star(P_i), d_\star(P_{i+1})] : d_\star \in \mathcal{D}\}\\
&\quad \cup \{[d_{\text{abs}}(P_i), d_\star(P_{i+1})] : d_\star \in \mathcal{D}_{\text{aux}}\},\\
\mathcal{I}_8 &= [d_\star(P_{i,1}), \ldots, d_\star(P_{i,8})], \qquad d_\star \in \mathcal{D},\\
\mathcal{I}_{\text{def}} &= [d_{\text{abs}}(P_i), d_{\text{abs}}(P_{i+1})].
\end{aligned}$$

Table 1: Model specifications for the reported machine-learning experiments.

Experiment	**Input**	**Shape**	**Arch.**	**Output head**	**Loss**	**Supervision target**
AESPTv2 CNN	ROI	101×1×1	A1	9-way softmax	Categorical CE	$\mathrm{HW}(S(P_{16} \oplus K_{16}))$
AESPTv2 LSTM	ROI	101×1	A2	9-way softmax	Categorical CE	$\mathrm{HW}(S(P_{16} \oplus K_{16}))$
Sec. 7.1 single-pair	$\mathcal{I}_1$	65×1	A3(1)	8 sigmoid outputs	Binary CE	8 bits of target byte y
Sec. 7.1 intra-pair	$\mathcal{I}_{\text{intra}}$	65×2	A3(2)	8 sigmoid outputs	Binary CE	8 bits of target byte y
Sec. 7.1 raw control	$\mathcal{I}_{\text{raw}}$	65×2	A3(2)	8 sigmoid outputs	Binary CE	8 bits of target byte y
Sec. 7.2 inter-pair	$\mathcal{I}_{\text{inter}}$	65×2	A3(2)	8 sigmoid outputs	Binary CE	8 bits of target byte y
Sec. 7.3 baseline	$\mathcal{I}_8$	65×8	A3(8)	8 sigmoid outputs	Binary CE	8 bits of Δ_{base}
Sec. 7.3 XOR-aware	$\mathcal{I}_8$	65×8	A4	8×256 logits	Sparse categorical CE (from logits)	$\Delta_c = \Delta_{\text{base}} \oplus o_c$
Sec. 8 countermeasures	$\mathcal{I}_{\text{def}}$	60×2	A3(2)	8 sigmoid outputs	Binary CE	8 bits of Δ_{base}

For the ARM bitwise models, the eight sigmoid outputs are interpreted as Bernoulli probabilities for the eight target bits. For each candidate byte, the corresponding bit probabilities are combined into a byte-level score, and scores are accumulated across attack examples in the log domain before GE/SR evaluation. The repeated GE/SR trials are randomized attack-subset resampling trials, not independently retrained networks. Optional post-training polarity calibration implemented in some scripts was not used for the reported results.

For the XOR-aware model, each of the eight channels produces 256 local-delta logits, which are remapped to the base-delta space using the known XOR offsets before accumulation. For the Sec. 8 countermeasure experiments, a fresh instance of the attacker is trained on profiling traces collected from the corresponding protected implementation and evaluated on a separate protected attack set.

For AESPTv2, each key-byte hypothesis is mapped to its predicted Hamming-weight class, and the corresponding class probability is accumulated across attack traces in the log domain to rank all 256 hypotheses.

Training and preprocessing protocols.

P1 – AESPTv2. The AESPTv2 D1 unprotected profiling corpus contains 100,000 traces. The corpus is shuffled with seed 42 and split into 80,000 training and 20,000 validation traces. Both SMOTE and non-SMOTE variants are evaluated; SMOTE augmentation is applied only to the training partition.

T1 – AESPTv2 CNN. Adam, learning rate 10^{-3}, batch size 128, 175 epochs, validation frequency 5.

T2 – AESPTv2 LSTM. Adam, learning rate 10^{-3}, batch size 128, 100 epochs, validation frequency 5.

P2 – ARM. The 3,000,000-trace ARM corpus is shuffled with seed 42 and split into 80% training and 20% validation subsets. Per-position and per-channel z-score normalization is fitted on the training partition and applied unchanged to validation and attack data. SMOTE is not used.

T3 – ARM CNNs. Adam, learning rate 10^{-3}, batch size 256, maximum 50 epochs; ReduceLROnPlateau with factor 0.5, patience 2, and minimum learning rate 10^{-5}; EarlyStopping with patience 5 and restoration of the best validation weights.

P3 – Protected ARM. The same preprocessing as P2 is applied to newly collected profiling traces from the corresponding protected implementation.

Table 2: Training and attack-evaluation settings for the reported machine-learning experiments.

Model family	**ROI**	**Prep.**	**Train**	**Attack evaluation**	**Independent trainings**
AESPTv2 CNN	1150–1250	P1	T1	600 traces; 256 key hypotheses; 100 randomized subsets per budget	1 per variant
AESPTv2 LSTM	1150–1250	P1	T2	300 traces; 256 key hypotheses; 100 randomized subsets per budget	1 per variant
Secs. 7.1–7.2	80–144	P2	T3	Up to 2,000 attack examples; 170 randomized subsets per budget	1 per configuration
Sec. 7.3 baseline	80–144	P2	T3	Up to 1,000 attack examples; 170 randomized subsets per budget	1 per configuration
Sec. 7.3 XOR-aware	80–144	P2	T3	Up to 2,000 attack examples; 170 randomized subsets per budget	1 per configuration
Sec. 8 cyclic shift	520–579	P3	T3	All 256 Δ_{base} values; up to 5,000 attack examples per value; 100 randomized subsets per budget	1 protected model
Sec. 8 coded index	520–579	P3	T3	Separate protected attack set; up to 10,000 attack examples; 170 randomized subsets per budget	1 protected model